On-site
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linkedin.com/in/nathaniel-vazquez-b41631153
quantumv.dev
steady-watch.com
kaizendoku.com

Nathaniel Vazquez

SOFTWARE ENGINEER

SUMMARY

Software engineer with a foundation in semiconductor manufacturing, enterprise operations, and hands-on education, now building at the intersection of quantum computing and applied cryptography. Comfortable operating across the full stack, from low-level hardware and sensor data to backend API architecture and polished mobile UI. Background in data-driven operations, team leadership, and technical instruction informs an engineering style that is systematic, well-documented, and built to last.

EXPERIENCE

CNC Operator

April 2026 - September 2026

American Manufacturing & Embroidery · On-site

- Operated multiple CNC cutting systems, including CAM5 CO2 laser, ATOM FlashCut pneumatic, and Thermwood 5-axis rotary platforms, to produce precision-cut components from shearling, premium leather, and liner materials.
- Set up and calibrated laser power, speed, and focus, and pneumatic die-cutting tooling, to hold tight tolerances and minimize scrap across production runs on high-value hides.
- Read and nested cut patterns to maximize material yield, and performed routine maintenance and safety checks on all equipment to maintain consistent uptime and part quality.

Senior Software Developer & Teaching Assistant

2022 - 2025

Persevere NOW · On-site

- Served as Senior Software Developer and full-time Teaching Assistant for multiple cohorts of 20 students, teaching software engineering fundamentals to underserved communities.
- Designed and delivered curriculum covering web development, programming fundamentals, and software engineering best practices; adapted instruction to diverse learning backgrounds and skill levels.
- Mentored students through project development from concept to deployment; provided code reviews, debugging support, and career guidance for aspiring developers entering the workforce.

Lead Pharmacy Technician, DOD Overseas & Expatriate Team

2012 - 2018

Express Scripts · On-site / Enterprise

- Led a team of ~20 CPhT-licensed professionals providing pharmaceutical services to deployed Department of Defense personnel and expatriate staff overseas; managed scheduling, quality control, and performance development.
- Drove data collection and analysis workflows to monitor prescription throughput, error rates, and compliance metrics across a high-volume mail-order fulfillment operation.
- Role evolved into enterprise-level management covering the full scope of overseas mail-order pharmaceutical services, coordinating logistics, regulatory compliance, and cross-functional team operations at scale.

Process Engineer, Test Wafer Distribution

2005 - 2011

Freescale Semiconductor · On-site

- Operated, tested, and maintained CVD, Cleans, Diffusion, EPI, and Photolithography equipment in the design and manufacture of microprocessors within a semiconductor fabrication environment.
- Role specialized into the management, data collection/analysis, and deployment of test silicon across the entirety of the manufacturing process, tracking wafer performance metrics and identifying process deviations.
- Developed hands-on expertise in precision manufacturing, process control, and hardware-level data analysis that directly informs current work in quantum hardware and cryptographic systems.

TECHNICAL SKILLS

LANGUAGES

Swift · SwiftUI · Python · JavaScript · HTML/CSS

MOBILE

iOS · watchOS · React Native · Expo · CoreMotion · HealthKit · WatchConnectivity · LocalAuthentication · CryptoKit · Secure Enclave

BACKEND

Python · Flask · Node.js · Express · Socket.IO · REST API design · JSON · SQLite

QUANTUM COMPUTING

Qiskit · IBM Quantum hardware · AerSimulator · GHZ circuits · QKD protocols · Mermin inequality

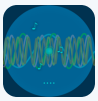
CRYPTOGRAPHY

P256 ECDSA · SHA-256 · Post-quantum encryption · Secure Enclave key management · Biometric key derivation

TOOLS & PRACTICES

Xcode · Git · GitHub · LaTeX · Agile · Technical writing · Patent documentation

NOTABLE WORK



STEADYWATCH HANDHELD | App Store v2.1

Shipped iOS + watchOS practice panel for musicians in Swift/SwiftUI: play-along tracks analyzed on-device for key, chords, tempo and bars, time-stretch without pitch change and transposition, on-beat recording, chord pairings on the circle of fifths, an atomic-time metronome, a harmonic tuner, and an Apple Watch remote, with a one-time Pro in-app purchase.



KaizenDoku | App Store v2.5.0

Shipped React Native / Expo puzzle app pairing Sudoku with cognitive wellness coaching: HealthKit sleep-performance correlation engine, real-time multiplayer (Socket.IO), Game Center leaderboards, RevenueCat in-app purchases, and an MCP-driven intelligent ad-placement system.



Quantum V^ Platform | quantumv.dev

Built and operate an entropy-as-a-service platform: a public QRNG API (Python/Flask on a Raspberry Pi behind a Cloudflare Tunnel) serving random bytes from entropy pools seeded on IBM Quantum hardware; QES-256, 32-byte keys traceable to verifiable IBM Quantum job IDs, with a drop-in replacement for Math.random(); and QES-KX, a hybrid X25519 + ML-KEM-768 (FIPS 203) key exchange. Live status, documentation and published research at quantumv.dev.